

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester III Examination, 2022-23

Paper: GLB-301 — Petrology and Economic Geology

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 055019

*Instructions: Answer **five** questions in all, selecting at least **two** each from Sections A and B, and Section-C which is compulsory.*

SECTION — A

Question 1. Write in brief about the classification of igneous rocks. [14]

Question 2. Discuss about primary sedimentary structures with suitable sketch. [14]

Question 3. Briefly describe the types of metamorphism. Discuss the principal controlling agents of different types of metamorphism. [14]

Question 4. Write short notes on **any four** of the following: $[4 \times 3\frac{1}{2} = 14]$

- (a) Porphyritic texture
- (b) Dyke and Sill
- (c) Limestone
- (d) Sandstone
- (e) Metamorphic zones
- (f) Isogrades

SECTION — B

Question 5. Write important chemical formula, distribution in India and uses of **any four** of the following economic minerals: $[4 \times 3\frac{1}{2} = 14]$

- (a) Baryte
- (b) Corundum
- (c) Chalcopyrite
- (d) Gypsum
- (e) Talc
- (f) Coal

Question 6. Write in brief **any two** of the following: [2 × 7 = 14]

- (a) What is coal? Discuss about the origin and economic uses of coal.
- (b) What are refractory minerals? Discuss about the economic uses of refractory minerals and their distribution in India.
- (c) What is chromite? Describe about economic uses and distribution of chromite ore deposits in India.

Question 7. Briefly describe about classification of economic mineral deposits. [14]

SECTION — C

Question 8. Answer the following:

- (a) Write the names of minerals formed in discontinuous Bowen's reaction series. [2]
- (b) Differentiate between breccia and conglomerate. [2]
- (c) Define cementation of sedimentary rocks. [1]
- (d) What is protolith? [1]
- (e) Write the name of a high grade metamorphic rock. [1]
- (f) Write the names of two petroleum deposits in India. [2]
- (g) Write the name of one ore mineral of aluminium. [1]
- (h) How many sets of cleavage planes are there in fluorite? [1]
- (i) Which metal is extracted from pyrolusite? [1]
- (j) What are the economic uses of galena? [2]